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جامعة عين شمس
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قسم نظم حاسبات

ECG Personal Photo Lock

HCI Project Report

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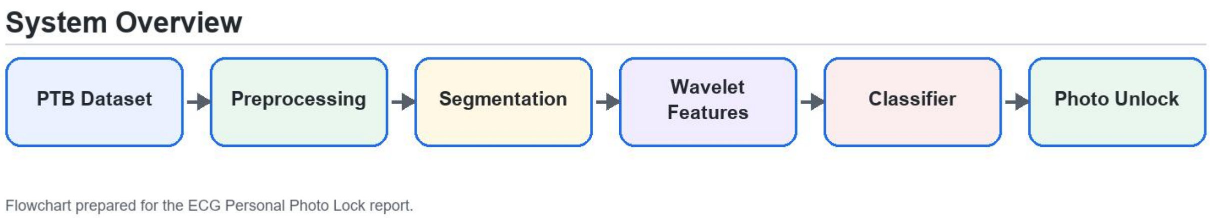


Figure 1. Overall ECG photo lock workflow.

1. Data Preparation

The data source is the PTB Diagnostic ECG Database from PhysioNet. Five subject folders are selected from the PTB patient records. In the current testing setup, the selected subjects are patient068, patient078, patient100, patient120, and patient140.

Each ECG record is read from WFDB format. Lead II is used for feature generation because it provides a clear ECG morphology and is commonly useful for R-peak detection. The extracted heartbeat feature rows are split into training and testing sets using a stratified split so each subject remains represented in both sets.

Database	PTB Diagnostic ECG Database
Subjects	Five PTB patients selected in the Training tab
Lead	Lead II, zero-based lead index 1
Sampling Frequency	1000 Hz
Train/Test Split	70% training, 30% testing

2. ECG Preprocessing

The preprocessing stage reduces noise and prepares stable heartbeat windows for classification. The pipeline removes baseline wander, applies bandpass filtering, suppresses powerline noise, detects R-peaks, and cuts fixed-length heartbeat segments around each detected R-peak.

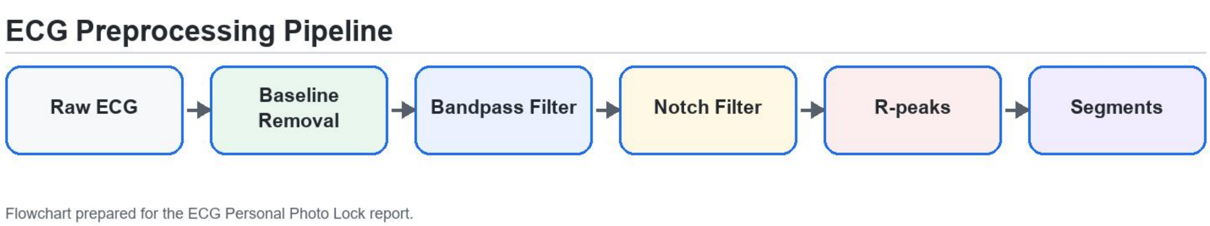
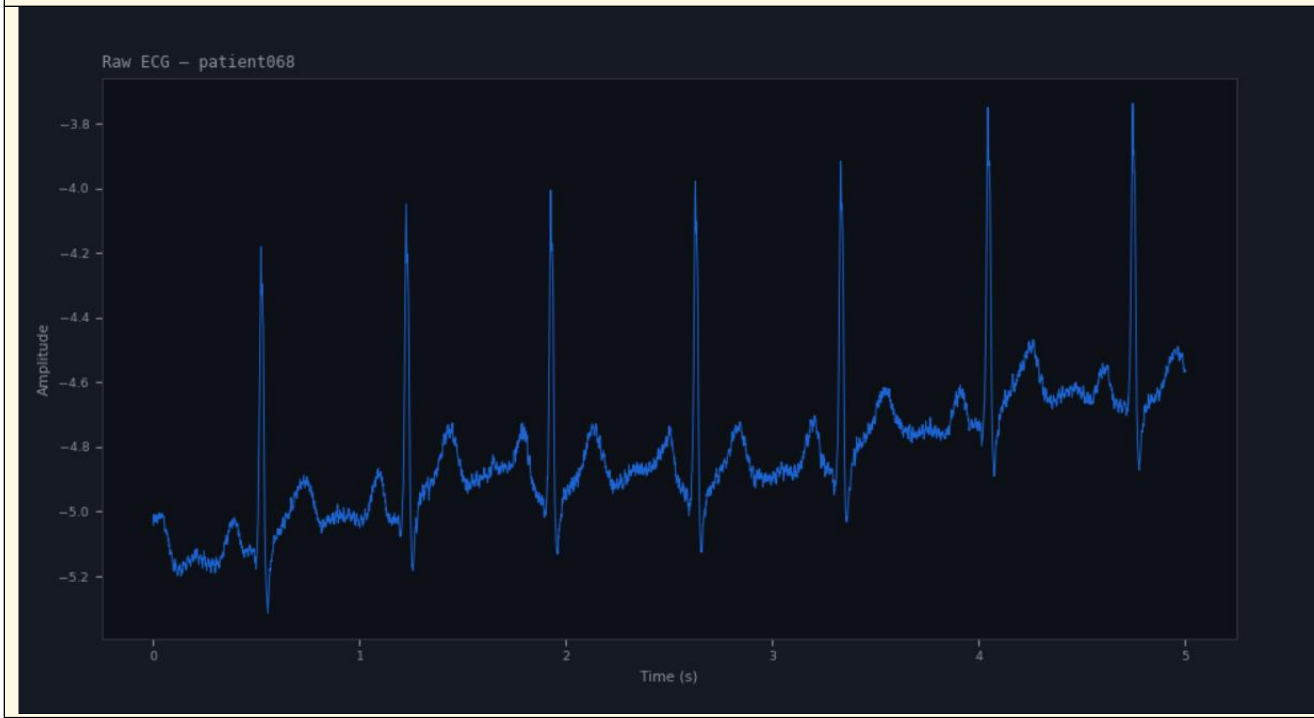


Figure 2. ECG preprocessing pipeline.

Raw ECG signal from ECG Viewer tab



Filtered ECG with detected R-peaks

3. Feature Extraction

The feature extraction method uses Discrete Wavelet Transform features. Three Daubechies mother wavelets are compared: db1, db2, and db4. For each heartbeat segment, wavelet coefficients are generated and summarized with statistical descriptors.

Mother Wavelets	db1, db2, db4
Transform Type	Discrete Wavelet Transform
Decomposition Level	Level 4
Per-band Features	Mean, standard deviation, energy, maximum absolute value, minimum absolute value
Purpose	Represent each ECG heartbeat segment numerically for classifier training
Comparison	The app trains and reports classifier performance separately for each wavelet

4. Classifiers and Parameters

Three classifiers are trained and compared. SVM is included as required, and KNN and Random Forest are used as additional classifiers. Several parameters are tested for each classifier to find the best result.

Classifier	Parameters Tested	Role in Comparison
SVM	Kernel: RBF, linear, polynomial; C: 1, 10, 100; gamma: scale or auto	Primary required classifier
KNN	Neighbors: 3, 5, 7, 11; metric: Euclidean or Manhattan	Distance-based baseline
Random Forest	Trees: 50, 100, 200; max depth: None, 10, 20	Tree ensemble baseline

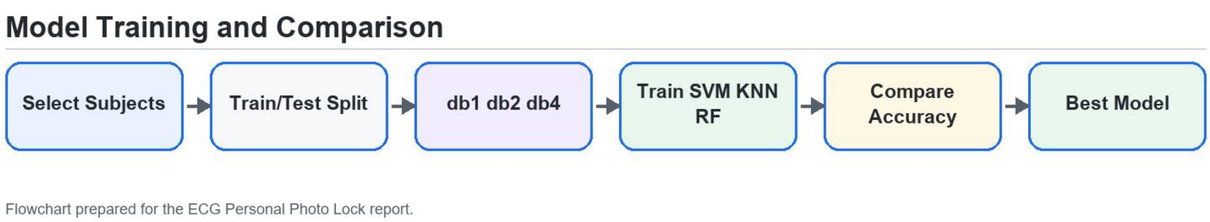
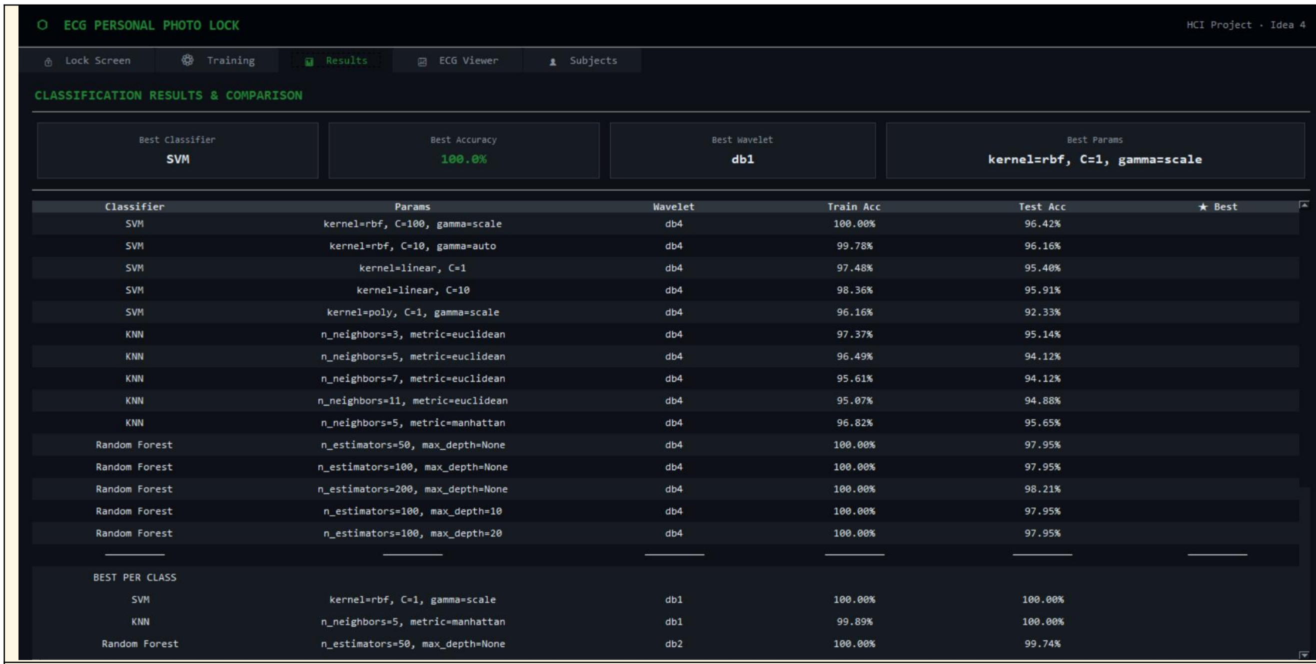


Figure 3. Training and model comparison workflow.

5. Classification Results

The table below summarizes one verified training run using the selected five patients: patient068, patient078, patient100, patient120, and patient140. If the final demo is retrained with another set of patients, replace the values with the final Results tab output.

Classifier	Best Wavelet	Best Parameters	Train Accuracy	Test Accuracy
SVM	db1	RBF kernel, C=1, gamma=scale	100.00%	100.00%
KNN	db1	k=5, Manhattan metric	99.89%	100.00%
Random Forest	db2	50 trees, max depth=None	100.00%	99.74%
Overall Best	db1	SVM, RBF kernel, C=1, gamma=scale	100.00%	100.00%



Classifier	Params	Wavelet	Train Acc	Test Acc	★ Best
SVM	kernel=rbf, C=100, gamma=scale	db4	100.00%	96.42%	
SVM	kernel=rbf, C=10, gamma=auto	db4	99.78%	96.16%	
SVM	kernel=linear, C=1	db4	97.48%	95.40%	
SVM	kernel=linear, C=10	db4	98.36%	95.91%	
SVM	kernel=poly, C=1, gamma=scale	db4	96.16%	92.33%	
KNN	n_neighbors=3, metric=euclidean	db4	97.37%	95.14%	
KNN	n_neighbors=5, metric=euclidean	db4	96.49%	94.12%	
KNN	n_neighbors=7, metric=euclidean	db4	95.61%	94.12%	
KNN	n_neighbors=11, metric=euclidean	db4	95.07%	94.88%	
KNN	n_neighbors=5, metric=manhattan	db4	96.82%	95.65%	
Random Forest	n_estimators=50, max_depth=None	db4	100.00%	97.95%	
Random Forest	n_estimators=100, max_depth=None	db4	100.00%	97.95%	
Random Forest	n_estimators=200, max_depth=None	db4	100.00%	98.21%	
Random Forest	n_estimators=100, max_depth=10	db4	100.00%	97.95%	
Random Forest	n_estimators=100, max_depth=20	db4	100.00%	97.95%	
———	———	———	———	———	———
BEST PER CLASS					
SVM	kernel=rbf, C=1, gamma=scale	db1	100.00%	100.00%	
KNN	n_neighbors=5, metric=manhattan	db1	99.89%	100.00%	
Random Forest	n_estimators=50, max_depth=None	db2	100.00%	99.74%	

Results Comparison

6. Authentication Interface

The Lock Screen provides the authentication interface. After training, the user can scan a his ECG signal or load an ECG file. The application preprocesses the signal, extracts wavelet features using the best model's wavelet window, predicts each heartbeat segment, and applies the majority-vote rule.

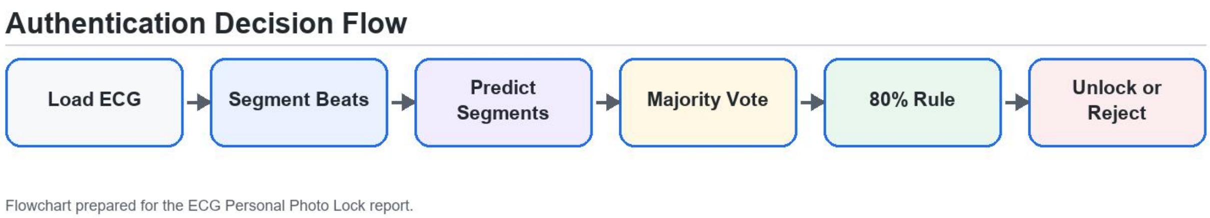
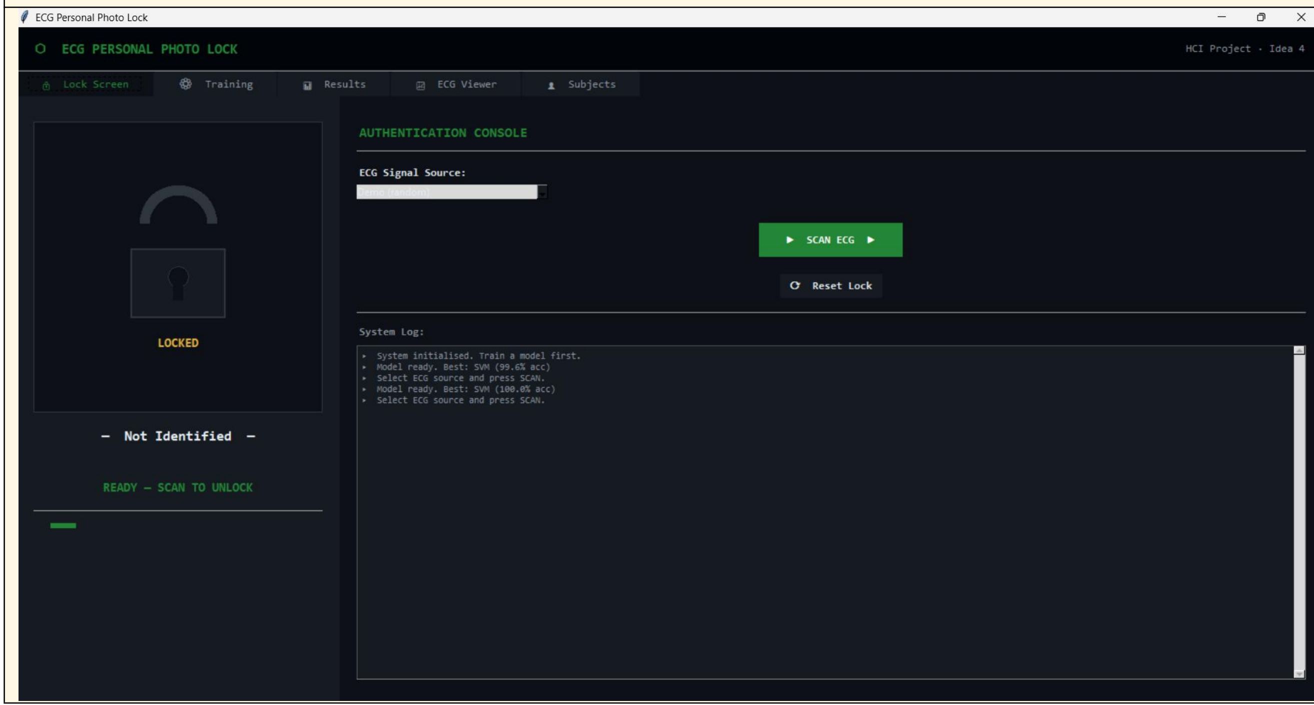


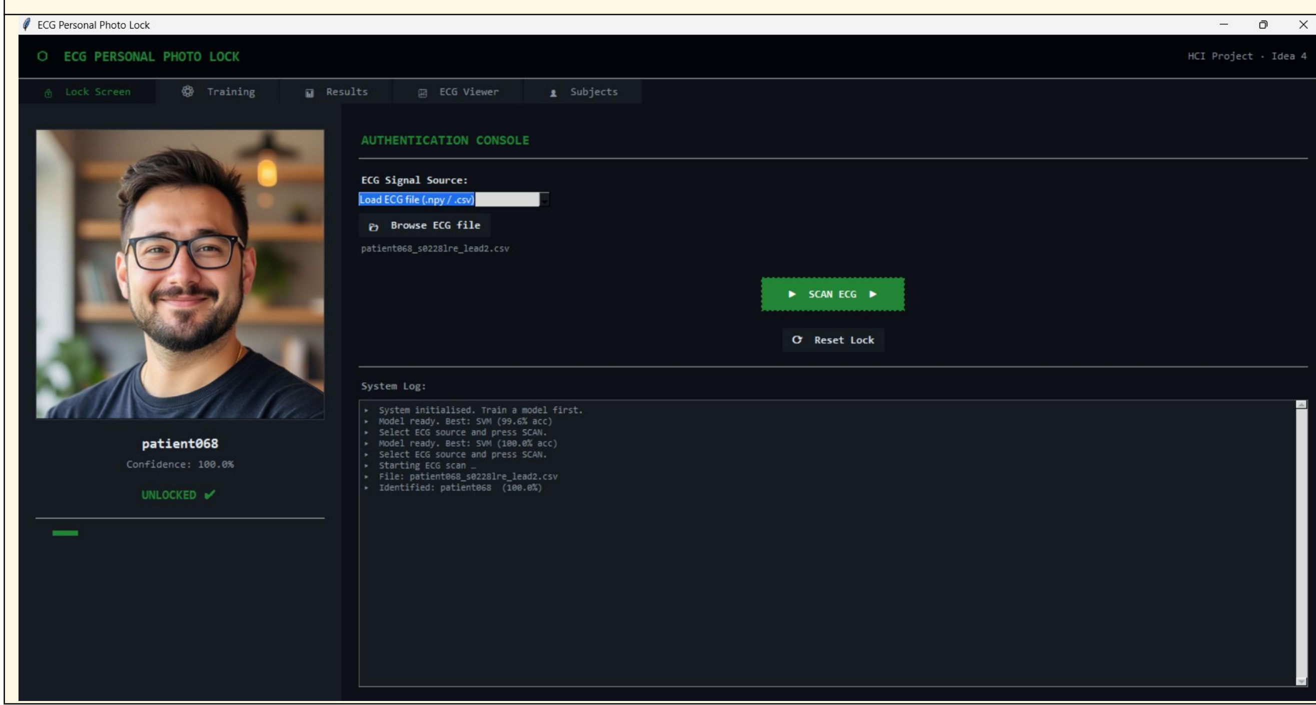
Figure 4. Authentication and unlock decision workflow.

When a known subject is identified, the matching subject photo is displayed. If the ECG belongs to a patient outside the trained five subjects, or if the confidence threshold is not met, access is denied and the subject is labeled unidentified.

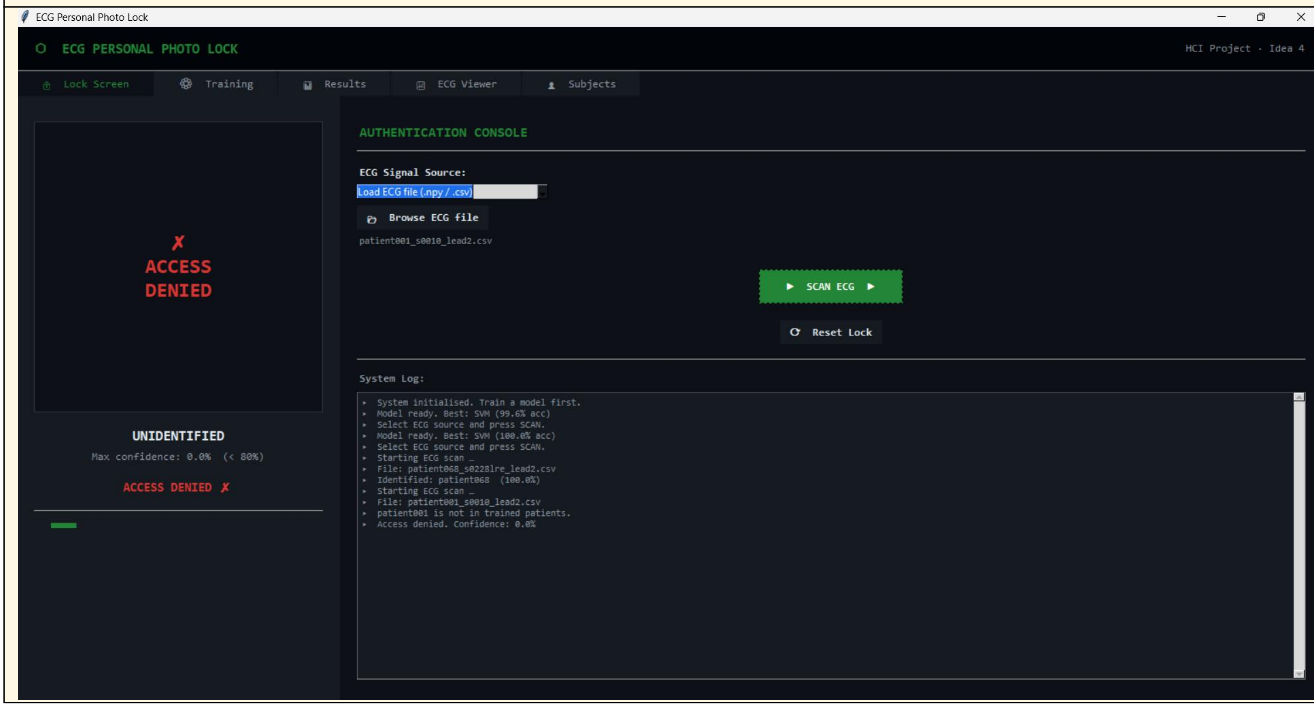
Insert screenshot: Lock Screen before scan



Insert screenshot: successful unlock with subject photo



Insert screenshot: unidentified / access denied case



Insert screenshot: Subject Manager with assigned photos